

Data Card — Lesotho Agricultural Advisory Small Language Model

1. Input vs. Output

- User inputs (text, in Sesotho or English):
- Short typed or voice-transcribed questions from farmers (e.g., planting dates, pest/disease symptoms, fertilizer guidance, livestock care)
- Optional structured inputs: location (district/elevation zone), crop type, growth stage, recent rainfall observations
- Model outputs:
- Plain-language advisory answers in the farmer's language (Sesotho or English)
- Each answer accompanied by: (a) a confidence indicator, (b) the source(s) the answer draws on, and (c) a recommendation to consult a local extension officer when confidence is low or the question involves regulated inputs (pesticides, veterinary medicine)

2. Classification Task

The model's core safety mechanism is a classifier that routes each incoming question before generation:

Class	Examples	Why it matters
Planting & seasonal timing	"When do I plant sorghum in the lowlands?"	High-stakes; answers tied to seasonal forecasts
Pest & disease identification	"What's eating my bean leaves?"	Risk of harmful misidentification; needs image/text triage
Soil & fertilizer	"Should I add urea to this field?"	Costly mistakes; regulated inputs
Livestock & grazing	"My cow won't eat, what do I do?"	Animal health; vet-escalation triggers
Weather & climate risk	"Will there be frost next week?"	Must cite Met Service data only
Out-of-scope / dangerous	Medical questions for humans, political topics, pesticide dosage prescriptions	Routed to refusal/escalation message

This taxonomy mirrors how extension officers categorize queries, keeping the model aligned with real agricultural workflows.

3. Labeling

Where labels come from (layered, with humans always in the loop):

- **Domain experts:** Lesotho Ministry of Agriculture extension officers and agronomists label the gold-standard answer set and validate sample outputs.
- **Community members:** Farmers co-annotate real question-answer pairs during field sessions; their corrections capture local knowledge (indigenous planting indicators, Sesotho crop terminology) that experts alone would miss.
- **AI-assisted drafting only:** LLMs may suggest draft annotations or translations, but every AI-assisted label is reviewed by a human reviewer fluent in Sesotho before it enters the dataset.

Consistency and accuracy checks:

- Two independent annotators label a 20% overlap sample; disagreements are resolved in a facilitated session, and the resolved rule is added to the annotation guide.
- Periodic "back-translation" spot checks: Sesotho answers re-translated to English by a third party to catch meaning drift.
- Extension officers audit a random monthly sample of final answers for agronomic correctness.

4. Data Creation

Primary sources (direct collection):

- Structured interviews and question-collection sessions with farmers across all four agro-ecological zones of Lesotho (lowlands, foothills, mountains, Senqu River Valley) and across gender/age groups
- Existing extension officer FAQ logs and Ministry of Agriculture advisory leaflets (with permission and attribution)
- Historical planting calendars and weather station records from the Lesotho Meteorological Services
- Translated and validated agronomy guidance from regional bodies (SADC, FAO Lesotho programmes)

Synthesis in low-resource settings:

- Since Sesotho agricultural text data is scarce, we will synthesize question variants (paraphrases in Sesotho dialectal variants, common misspellings from feature-phone typing) from a small verified seed set — but all synthetic data is clearly flagged in the dataset and used only for training data augmentation, never presented as real farmer data.
- Back-translation (English→Sesotho→English) to expand coverage, with native-speaker validation of the final Sesotho output only.

5. Web Collection Ethics

- **Minimal web scraping.** We will draw primarily on official, openly licensed sources (FAO, CGIAR centres, government publications, Creative Commons agronomy resources) rather than scraping forums or social media.
- **No collection from social media or personal pages** — posts by Basotho farmers may contain personal details and were not posted for AI training; using them would violate the consent expectations of the posters.
- **Tools:** Human researchers manually collect and curate; AI tools assist with deduplication and formatting, not collection decisions.
- **Consent & privacy:** Interview participants give informed consent in Sesotho (oral consent option for low-literacy participants), with the right to withdraw their data at any time. Names, phone numbers, and precise farm locations are stripped before data enters the corpus; only district-level geographic tags are retained.
- **Copyright:** Each document is logged with its license and source; anything without a clear license or permission is excluded.

6. Bias and Representation

Who might be left out — and how we prevent it:

At-risk group	Risk	Mitigation
Women farmers	Underrepresented in extension systems; their questions differ (household gardens, poultry, post-harvest)	Targeted recruitment: ≥50% of interview participants are women; separate women's focus groups so male voices don't dominate
Older farmers	Less digitally connected; hold deep local ecological knowledge	Oral data-collection sessions; their knowledge is a priority input, not an afterthought
Mountain/remote district farmers	Hard to reach; very different climate from lowlands	Deliberate sampling quotas per agro-ecological zone, including Mokhotlong and Thaba-Tseka
Sesotho dialect speakers	Urban Sesotho dominates written text	Recruit annotators from rural areas; document dialect variants in the annotation guide
Non-smartphone users	Invisible in app analytics	Design data collection around the extension officer channel too, so offline farmers are represented

Ongoing check: Publish a demographic and geographic distribution report with each dataset release so exclusion is visible, not hidden.

7. Values and Community Alignment

How the dataset reflects our values:

- **Accessibility & linguistic relevance:** Sesotho is a first-class dataset language, not a translation afterthought. Farmers contribute data in their own words.
- **Accuracy & trustworthiness:** Every training answer traces to a verified source; the "I don't know / ask an extension officer" class is deliberately represented in training data so the model learns humility.
- **Community ownership:** Farmers and extension officers are named contributors (with consent) and sit on the dataset review board. A community advisory panel — including farmer representatives — reviews dataset releases and can request removal of data.

How the community benefits:

- Free, timely advice in their language that previously required travel to an extension office
- Extension officers gain a triage tool, multiplying their reach in under-staffed districts
- Sesotho agricultural vocabulary and local knowledge are preserved and strengthened in digital form rather than extracted without credit

Accountability mechanisms:

- Publicly released Data Card (like this one) with every dataset version
- A grievance channel: farmers can report wrong or harmful advice via SMS
- Annual community review session where findings — including failures — are reported back to participating districts in Sesotho